



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2022 Dec: IIT M DIPLOMA AN1 EXAM ETD1

Exam Mode

Learning Mode

View Question Paper Summary

QUESTION MENU

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CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 640653455091

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**FOUNDATION LEVEL/DIRECT ENTRY DIPLOMA: MATHEMATICS FOR DATA SCIENCE 2**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT ? CROSS CHECK YOUR HALL TICKET TO

CONFIRM THE SUBJECT TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT ,
PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Your score : 0

Discussions (0)



Question 2 : 640653455092

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true?



OPTIONS :

☐ If a system of linear equations $Ax = b$ has no solution, then b is a linear combination of the columns of the coefficient matrix A .



☐ Consider a system of linear equations $Ax = b$. If b is a linear combination of the columns of the coefficient matrix A , then the system $Ax = b$ has a solution.



☐ If the number of variables is more than the number of equations in a system of linear equations $Ax = 0$, then it has a unique solution.



☐ The kernel of the linear transformation $S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $S(v) = Av^T$, where $v = (x, y, z)$, is the solution space of the system of linear equations $Av^T = 0$.



Your score : 0

Discussions (0)



Question 3 : 640653455098

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following subspaces have a basis whose cardinality is 2?



OPTIONS :

☐ $\{A \in \mathcal{M}_{3 \times 3}(\mathbb{R}) : A \text{ is a diagonal matrix and the sum of the diagonal entries of } A \text{ is } 0\}.$

☐ $\{A \in \mathcal{M}_{3 \times 3}(\mathbb{R}) : A \text{ is a scalar matrix}\}.$

☐ $\text{span}\{(1, 3, 5), (2, 6, 10), (-3, -9, -15)\}.$

☐ $\text{span}\{(1, 1, 0, 0), (1, 1, 1, 0), (4, 4, 1, 0), (-1, -1, 5, 0)\}.$

Your score : 0

Discussions (0)



Question 4 : 640653455112

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider the affine subspaces $L_1 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 1\}$ and $L_2 = \{(x, y, z) \in \mathbb{R}^3 : x = 3\}$. Let W_1 and W_2 be the subspaces corresponding to L_1 and L_2 respectively. Choose the correct option(s).



OPTIONS :

☐ $W_1 = \text{span} \{(1, -1, 0), (0, 1, -1)\}.$

☐ $W_2 = \text{span} \{(1, 0, 0), (0, 1, 0)\}.$

☐ $W_1 \cap W_2$ represents a plane passing through the origin in $\mathbb{R}^3$.

☐ $W_1 \cap W_2$ represents a line passing through the origin in $\mathbb{R}^3$.

Your score : 0

Discussions (0)



Question 5 : 640653455111

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Let A be a 3×3 orthogonal matrix. Which of the following options is/are true?

OPTIONS :

☐ The rows of A are not linearly independent.☐ The system of linear equations $Ax = b$, where $b \neq 0$ has infinitely many solutions.☐ The columns of A form an orthogonal set of vectors of the inner product space $\mathbb{R}^3$ w.r.t. the dot product.☐ The rank of A is 3.

Your score : 0

Discussions (0)



Question 6 : 640653455097

View Solutions (0)

Total Mark : 3.00 | Type : SA

Consider $V = \mathbb{R}^3$ with inner product as the dot product and $W = \{(x, y, z) \mid x = y\}$ is a subspace of V . If (a, b, c) is the projection of $(1, 2, 3)$ onto W , then what is $a + b + 2c$?

Answer (Numeric):

Answer

Accepted Answer : 9

Your score : 0

Discussions (0)



Question 7 : 640653455093

Total Mark : 0.00 | Type : COMPREHENSION

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T(x, y) = (x + y, x - y)$.

Answer the given subquestions.

Your score : 0



Question 8 :

640653455094



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Choose the correct option(s) about T .



OPTIONS :



T is not onto.



The range of T is a one-dimensional subspace of $\mathbb{R}^2$.



The nullity of T is 0.



T is one-one.



Your score : 0

Discussions (0)



Question 9 :

640653455095



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the matrix representation



of T with respect to a basis $\{(1, 1), (1, 2)\}$

for the domain and $\{(1, 1), (1, -1)\}$

for the co-domain, then what is $b + c$?

Answer (Numeric):

Accepted Answer : 2

Your score : 0

 Discussions (0)

Question 10 :

640653455096



View Parent QN



View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let A be the matrix representation of T with respect to a basis $\{(1, 1), (1, 2)\}$ for the domain and $\{(1, 1), (1, -1)\}$ for the co-domain (obtained in the previous question) and B be the matrix representation of T with respect to the standard ordered basis for $\mathbb{R}^2$ for both the domain and co-domain. Choose the correct option(s).



OPTIONS :

☐ Rank of the matrix A is 2. ☐ Both the systems $Ax = 0$ and $Bx = 0$ have infinitely many solutions. ☐ A and B are similar matrices. ☐ A and B are equivalent matrices. 

Your score : 0

 Discussions (0)

Question 11 : 640653455099

Total Mark : 0.00 | Type : COMPREHENSION

Consider the function $f(x, y, z) = x^2 + y^2 + z^2 + xy + yz + kzx$. Use this information to answer the given subquestions.



Your score : 0



Question 12 :
640653455100

[View Parent QN](#)
[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Which of the following matrices represent the transpose of the Hessian matrix of the function at any critical point?



OPTIONS :

☐ $\begin{bmatrix} 2 & 1 & k \\ 1 & k & 1 \\ k & 1 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} 2 & 1 & k \\ 1 & 2 & 1 \\ 2 & 1 & k \end{bmatrix}$

☐ $\begin{bmatrix} 2 & 1 & k \\ 1 & 2 & 1 \\ k & 1 & 2 \end{bmatrix}$

☐ $\begin{bmatrix} k & 1 & 2 \\ 1 & k & 1 \\ 2 & 1 & k \end{bmatrix}$

Your score : 0

[Discussions \(0\)](#)**Question 13 :****640653455101**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

Which of the following options is/are true?

OPTIONS :

- ☐ If $k = 1$, then the function attains a value at the critical point which is locally maximum
- ☐ If $k = 2$, then the Hessian test is inconclusive.
- ☐ There is at least one value of k for which the critical point is a saddle point.
- ☐ If $k = 3$, then the function attains a value at the critical point which is locally minimum.

Your score : 0

[Discussions \(0\)](#)**Question 14 :****640653455102**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

Which of the following options is/are true?

OPTIONS :

- ☐ Equations $f_x = 0, f_y = 0, f_z = 0$ do not form a system of linear equations.
- ☐ For any value of k the Hessian matrix is a symmetric matrix.

- ☐ If $k = 2$, then the nullity of the Hessian matrix is 1. 
- ☐ For any value of k , the Hessian matrix has rank 2. 

Your score : 0

 Discussions (0)



Question 15 :
640653455103

 View Parent QN

 View Solutions (0)

Total Mark : 3.00 | Type : SA

Let H denote the matrix obtained by putting $k = 1$ in the Hessian matrix. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation $T(x, y, z) = (a_1x + b_1y + c_1z, a_2x + b_2y + c_2z, a_3x + b_3y + c_3z)$. If H is the matrix representation of T w.r.t the standard ordered basis $\{e_1, e_2, e_3\}$ for both domain and codomain, then find the value of $2(a_1 + a_2) + (b_2 + b_3) - (c_1 + c_3)$. 

Answer (Numeric):

Answer

Accepted Answer : 6

Your score : 0

 Discussions (0)



Question 16 : 640653455104

Total Mark : 0.00 | Type : COMPREHENSION

Consider the plane $x + y + z = 1$ in $\mathbb{R}^3$. Let (a, b, c) be the point on the plane such that the distance of the point from the origin is the least. 
Use this information to answer the given subquestions.

Your score : 0



Question 17 :
640653455105



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the value of $15(a + b)$. 

Answer (Numeric):

Answer

Accepted Answer : 10

Your score : 0

 Discussions (0)



Question 18 :
640653455106



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

If d is the minimum distance from the 
the origin, then find the value of $9d^2$.

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

 Discussions (0)



Question 19 : 640653455107

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the function $f(a, b) = b^3 + \int_0^a (3x^2 + 3b^2 - 15)dx$.



Your score : 0



Question 20 :
640653455108



View Parent QN



View Solutions (0)

Total Mark : 3.00 | Type : SA

Find the number of critical points.



Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

Discussions (0)



Question 21 :
640653455109



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Let $p_1 = (1, -2)$, $p_2 = (0, \sqrt{2})$, $p_3 = (\sqrt{5}, 0)$,
 $p_4 = (-1, 2)$, $p_5 = (-\sqrt{5}, 0)$ and



Answer the following questions
with respect to these points

Write down the values of i in increasing order
for which p_i , $i = 1, 2, 3, 4, 5$, is a
point of local minima according to the
Hessian test e.g. if p_1 and p_2 are points
of local minima, your answer
should be 13. If there is no such i ,
enter 0 as your answer.

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)



Question 22 :

640653455110



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Let $p_1 = (1, -2)$, $p_2 = (0, \sqrt{2})$, $p_3 = (\sqrt{5}, 0)$, $p_4 = (-1, 2)$, $p_5 = (-\sqrt{5}, 0)$ and

Answer the following questions

with respect to these points.

Write down the values of i in

increasing order for which

p_i , $i = 1, 2, 3, 4, 5$, is a saddle point

according to the Hessian test e.g.

if p_1 and p_3 are saddle points,

your answer should be 13.

If there is no such i , enter 0 as your answer.

Answer (Numeric):

Answer

Accepted Answer : 14

Your score : 0

Discussions (0)



Question 23 : 640653455113

Total Mark : 0.00 | Type : COMPREHENSION

Answer the given subquestions about the functions 

$$u(x, y) = x^2 + e^y \text{ and } v(x, y) = y^2 + ye^x$$

Your score : 0



Question 24 :
640653455114

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MSQ

If (s_1, s_2) is the unit vector along the direction in which the directional derivative of the function $u(x, y)$ at the point $(1, 1)$ is the maximum and (r_1, r_2) is the unit vector along the direction in which the directional derivative of the function $v(x, y)$ at the point $(1, 1)$ is the minimum, then which of the following options is/are true?



OPTIONS :

☐ $s_1 + s_2 = \frac{2+e}{\sqrt{4+e^2}}$ 

☐ $s_1 + s_2 = -\frac{2+e}{\sqrt{4+e^2}}$ 

☐ $r_1 + r_2 = \frac{2+2e}{\sqrt{2e^2+4e+4}}$ 

☐ $r_1 + r_2 = -\frac{2+2e}{\sqrt{2e^2+4e+4}}$ 

Your score : 0

[Discussions \(0\)](#)**Question 25 :**
640653455115[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If $L_u(x, y)$ is the linear approximation of the function $u(x, y)$ at point $(2, 3)$ and $L_u(3, 4) = (a + be^3)$, and if $L_v(x, y)$ is the linear of the function $v(x, y)$ at point $(1, 2)$, approximation and $L_v(3, 4) = (c + de)$, then find the value of $(a + c) - (b + d)$, where a, b, c, d are integers.



Answer (Numeric):

Accepted Answer : 10

Your score : 0

[Discussions \(0\)](#)**Question 26 :**
640653455116[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MSQ

Which of the following options is/are true?



OPTIONS :

☐ $z = 2x + e^2y - 1 - e^2$ is the tangent plane to the function $u(x, y)$ at point $(1, 2)$.

☐ $x(t) = 1 + \frac{1}{\sqrt{2}}t, y(t) = 2 + \frac{1}{\sqrt{2}}t,$
 $z(t) = (1 + e^2) + \frac{e^2}{\sqrt{2}}t$ is the tangent
 line of the function $u(x, y)$ at point
 $(1, 2)$ in the direction of $\frac{1}{\sqrt{2}}(1, 1)$.

☐ $z = 2ex + (4 + e)y + 4 - 3e$ is the tangent plane to the function
 $v(x, y)$ at point $(1, 2)$.

☐ $x(t) = 1 + \frac{1}{\sqrt{2}}t, y(t) = 2 + \frac{1}{\sqrt{2}}t, z(t) = (4 + 2e) + (2\sqrt{2} + \frac{3e}{\sqrt{2}})t$ is the
 tangent line of the function $v(x, y)$ at point $(1, 2)$ in the direction of $\frac{1}{\sqrt{2}}(1, 1)$.

Your score : 0

Discussions (0)



Question 27 :
 640653455117

View Parent QN

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider a function $g(x, y)$ such that

$$g(x, y) = \begin{cases} u(x, y) & x = y \\ v(x, y) & x \neq y \end{cases}$$

Which of the following options is/are true?

OPTIONS :

☐ $g(x, y)$ is continuous at the origin.

☐ $g_{xy}(1, 2) = e.$

☐ $g_{xy}(x, y)$ is not continuous at $(1, 2)$.

☐ $g_y(1, 1) = 2 + e.$

Your score : 0

 Discussions (0)



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